Sahaj Bhadja

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Summary

Software engineer with 2+ years building scalable backend services, distributed workflows, real-time applications, and data-intensive platforms using Python, APIs, cloud infrastructure, automated testing, and performance-focused system design.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant backend services on Azure SQL and Blob Storage for 500+ users, optimizing schemas, access patterns, and indexed queries to reduce dashboard latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workers for 1,000+ monthly jobs with explicit states, retries, and failure isolation, tripling concurrent throughput while preventing cascading failures across critical production workflows.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows handling 10,000+ daily requests, integrating secure file ingestion, automated evaluation, and model routing while maintaining stable performance under peak production load.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading and fault-tolerant execution, reducing turnaround by 60% while improving observability and overall operational reliability.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*
- Built a GPT-based Decision Transformer for autonomous-driving simulations, processing 100,000+ trajectory logs and improving long-horizon planning by 18% through reproducible data pipelines and systematic model evaluation.
- Architected a spatial-temporal encoder combining HD-map geometry and historical vehicle trajectories through multi-head attention, improving closed-loop planning stability by 15% across diverse, complex simulated urban driving scenarios.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites with Jenkins CI/CD, improving release reliability while supporting integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL components using OOP patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline efficiency by 68% overall.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex business requirements into maintainable real-time production workflows.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving implementation consistency, technical documentation, and issue resolution across frequent product updates.

Skills

Languages & Fundamentals: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, data structures, algorithms, object-oriented design, system design, concurrency, multithreading; **Backend & Data:** FastAPI, Flask, Node.js, REST APIs, WebSockets, microservices, distributed systems, PostgreSQL, MySQL, Azure SQL, MongoDB, ETL, real-time processing; **Cloud & Reliability:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, PyTest, Mockito, automated testing, debugging, monitoring, performance optimization, fault tolerance, Agile/Scrum

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, enabling performance comparison across 1,000+ simulated process combinations while validating preemptive and non-preemptive execution edge cases.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a scalable platform for 50,000+ users and 1M+ transactions, combining secure APIs, real-time synchronization, advanced filtering, and Kubernetes-backed deployments for reliable high-volume operational data workflows.

Trend Tracker Live | *Python, CNN, React.js, AWS EC2, Strapi*
- Developed a recommendation platform serving 3M+ users, improving model accuracy from 78% to 92% through CNN-based personalization while deploying responsive product experiences and scalable AWS-backed production services.